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Chapter

PSYCH 301
Cognitive Processes

Lecture 16:
Judgment, Decision, and Reasoning

Fall 2026

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Emotions

- People with damage to an area of their prefrontal cortex, who suffer from flattened emotions, have impaired decision making

- Anxious people tend to avoid making decisions that could potentially lead to large negative consequences
  - ➔ risk avoidance

- Optimistic people are more likely to ignore negative information and focus on positive information
  - Too much optimism can lead to poor decision making

Emotions

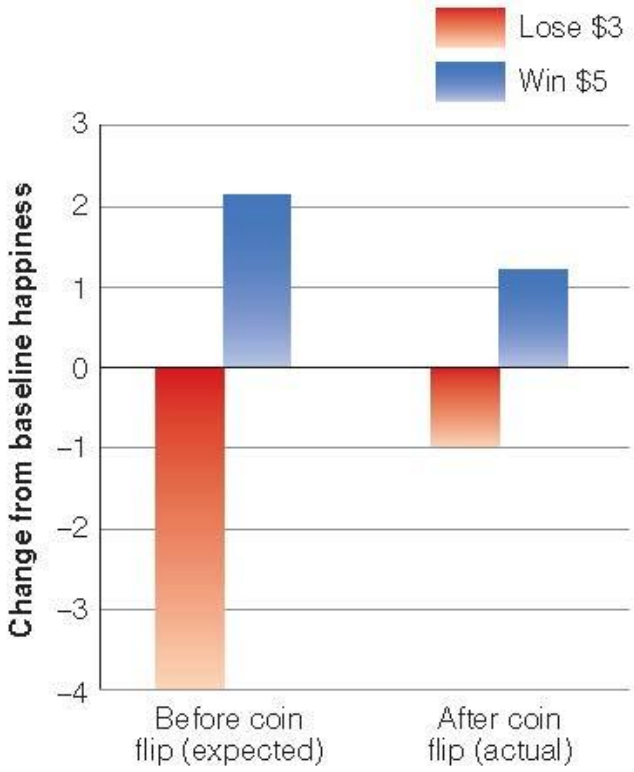
- Expected emotions: Emotions that people *predict* that they will feel concerning an outcome
- Expected emotions are one of the determinants of risk aversion (tendency to avoid taking risks)
- People inaccurately predict their emotions
- Tendency to believe that a particular loss will have a greater impact than a gain of the same size increases risk aversion

Decision-Making

Kermer et al. (2006): compared people's expected emotions with their actual emotions

•They gave S $5
•Told them that based on a coin flip they would either win an additional $5 or lose $3
•S rated their happiness before the experiment started
•They predicted how they think they would feel if they win or lose

•➔ overestimated their negative feeling
•We usually have coping mechanisms that get activated, but we don't consider them when predicting emotions



Emotions

- Incidental emotions: Emotions that are not specifically related to decision-making
  - May be related to one's general disposition or personality, some recent experience, or one's general environment or surroundings



- Analysis of university admissions decisions shows that applicants' academic attributes were more heavily weighted on cloudy days than on sunny days (Simonsohn, 2007)



- Prospective students visiting an academically highly rated university were more likely to enroll if they had visited the campus on a cloudy day (Simonsohn, 2009)

Emotions

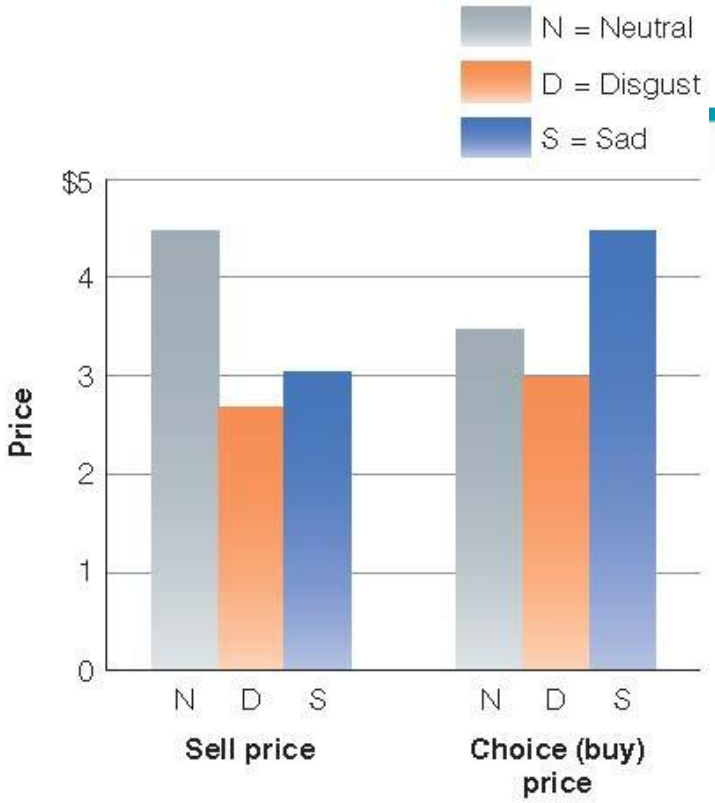
Lerner et al. (2004): S viewed one of these clips:

- A person dying (sad condition)
- A person using dirty toilet (disgust condition)
- Fish at the Great Barrier Reef (neutral)

•Next, a set of pens were given to S
•They were asked to determine the price

- they would be willing to sell the highlighters
- they would be willing to buy the highlighters

- ➔ a person's mood can influence decision making

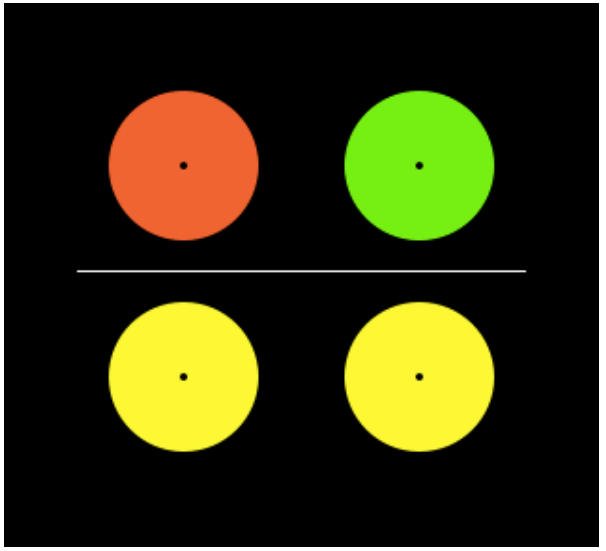




Context

- Decisions can depend on the context within which they are made

- Physicians were presented with a hypothetical test case involving a possible candidate for cesarean section. Decisions were made under three different context:
  - Control: the test case was presented first
  - Serious previous cases: test case was preceded by four other cases in which there were serious complications
  - Not serious previous cases: Test case was preceded by four other cases that were fairly routine
  - ➔ significantly more doctors recommended cesarean after reading about non-serious cases first



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Decision-Making



The International Community A chicken, a duck, a turkey, a pheasant, a quail, and a partridge were summoned and traveled together to the summit.

The king's chef welcomed them:

"I called you here," he explained, "so you can tell me what sauce you would like to be eaten with."

One of the birds dared to speak:

"I don't want to be eaten at all."

The chef quickly set it straight:

"That is out of the question," he barked.

- Decisions depend on how choices are presented
  - Opt-in procedure
    - Active step to be organ donor
    - Countries with opt-in procedures have less than 28% consent rate
  - Opt-out procedure
    - Organ donor unless request not to be
    - Countries with opt-our procedures have over 99% consent rate

Status quo bias

- The tendency to do nothing when faced with a difficult decision

Risk-Taking Strategies

- Framing effect: decisions are influenced by how a decision is stated
  - Can highlight one aspect of situation
    - Tversky and Kahnemann (1981)
    - When situations are framed in terms of gains, people tend toward a risk-aversion strategy
    - When situations are framed in terms of losses, people tend toward a risk-taking strategy

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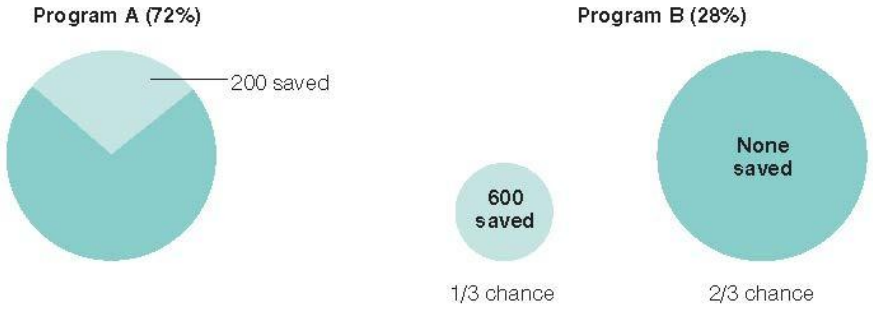
[Risky Decisions Lab]

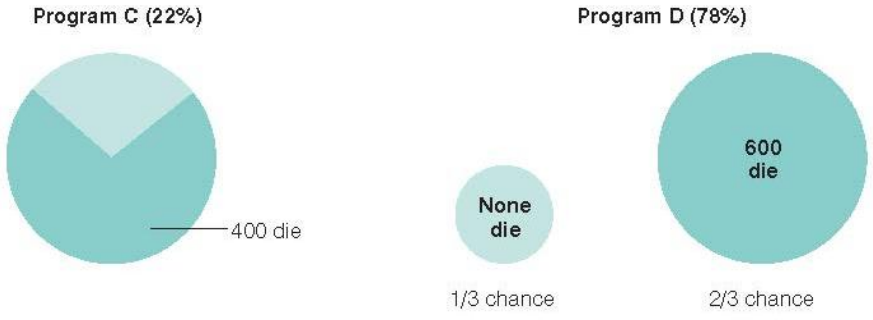
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Decision-Making

A Disease is expected to kill 600 people. Two alternative programs to combat the disease have been proposed

- Program A: 200 will be saved
- Program B: 1/3 probability that 600 people will be saved, and 2/3 probability that no people will be saved
- ➔ 72% chose A



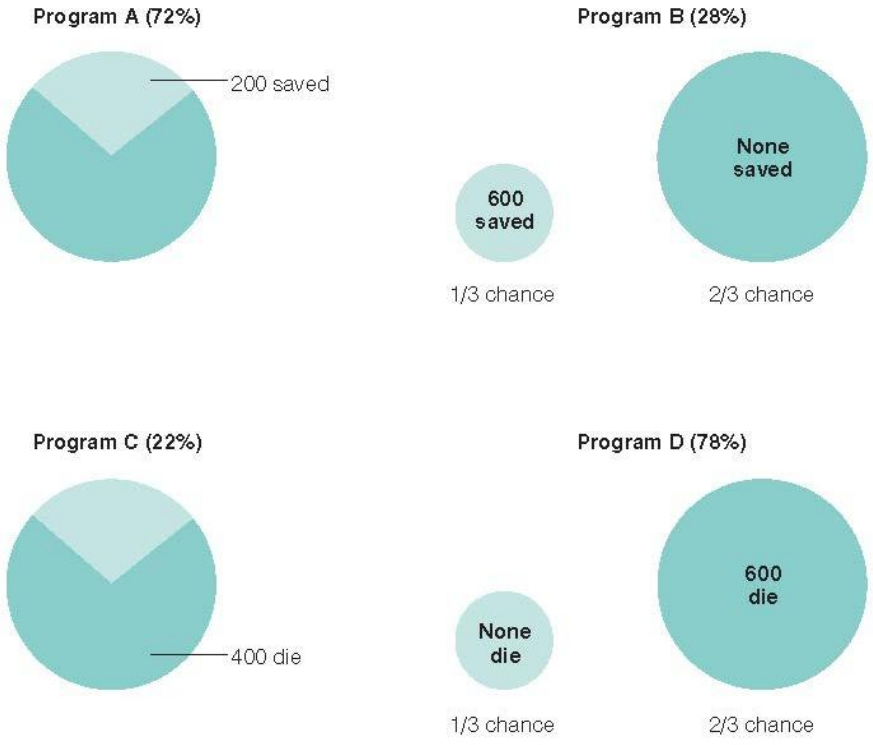


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Decision-Making

A Disease is expected to kill 600 people. Two alternative programs to combat the disease have been proposed

- Program C: 400 people will die
- Program D: 1/3 probability that no one will die, 2/3 probability that 600 people will die
- ➔ 78% chose D



The Physiology of Thinking

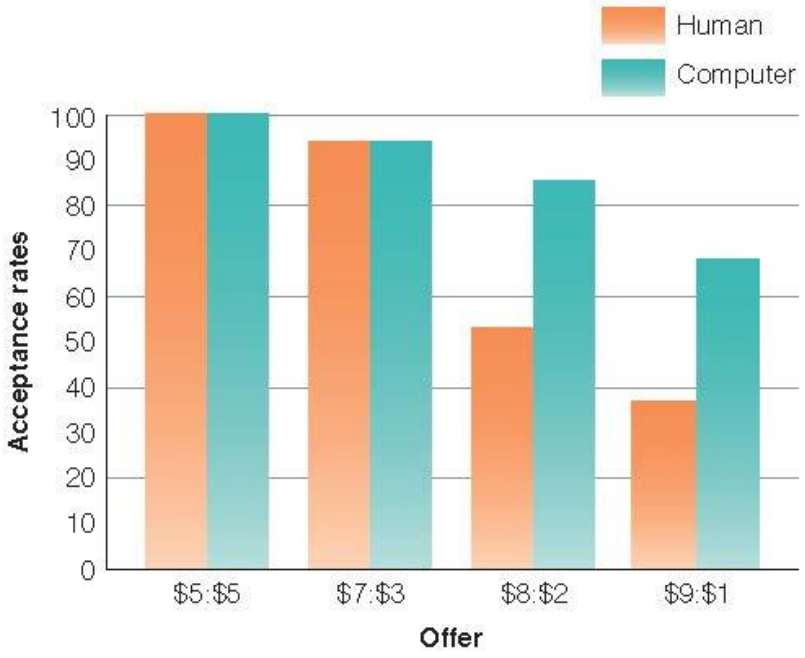
- Neuroeconomics
  - Combines research from psychology, neuroscience, and economics
  - One finding: decisions are influenced by emotions, and those emotions are associated with activity in specific areas of the brain

The Physiology of Thinking

- Sanfey and coworkers (2003)
  - Ultimatum game
    - Two players: proposer and responder
    - Proposer gets $10
    - Proposer makes an offer to the responder as to how this money should be split
    - Responder can accept or reject the offer.
    - If responder rejects, neither receive anything
    - According to utility theory, what is the rational decision for the responder?

The Physiology of Thinking

- In Sanfey's experiment, S play 20 separate games
- 10 with human partners, and 10 with computer partners
- Often rejected low offers because they became angry that offers were unfair
- Less angry with an "unfair" computer



Sanfey and coworkers (2003)

More activation of right anterior insula (connected with emotional states), participants more likely to reject more offers

Emotion is important in decision-making

Deductive Reasoning

- Deductive Reasoning: Determining whether a conclusion *logically follows* from premises

- Syllogism: basic form of deductive reasoning
  - Introduced by Aristotle
  - Two statements called premises
  - Third statement called conclusion

- Categorical syllogism
  - Describe relation between two categories using *all, no,* or *some*
    - *Premise 1: All birds are animals*
    - *Premise 2: All animals eat food*
    - *Conclusion: Therefore, all birds eat food*

Deductive Reasoning

- Syllogism is valid if conclusion follows *logically* from its two premises
- If two premises of a valid syllogism are true, the syllogism's conclusion must be true
  - Do not confuse "validity" with "truth"
    - *All birds are animals*
    - *All animals have four legs*
    - *All birds have four legs*
    - Valid! Same form as the first example
  - Validity is about whether the conclusion logically follows from the premises
  - If it does, and the premises are true, then the conclusions must be true

How Well Can People Judge Validity?

- The difference between validity and truth can make it difficult to judge whether reasoning is "logical" or not
  - Valid syllogisms can result in false conclusions
  - Syllogisms can be invalid even though each of the premises and the conclusion seem reasonable
    - *All of the students are tired*
    - *Some tired people are irritable*
    - *Some of the students are irritable*
    - Not valid!

- Many errors in evaluation: good reasoning and truth are not the same thing
  - Belief bias: The tendency to think that a syllogism is valid if its conclusions are believable
  - 71% of S judge invalid, believable conclusions as valid

Mental Model Approach

- There is no easy method for determining whether a syllogism is valid or invalid

  On a pool table there is a black ball directly above the cue ball. The green ball is on the right side of the cue ball, and there is a red ball between them. If I move so the red ball is between me and the black ball, the cue ball is to the _____ of my line of sight.

- Most people solve this by imagining the way the balls are arranges on the pool table

- The idea that people can imagine situations is the basis of Johnson-Laird's proposal that people use mental models to solve deductive reasoning

Mental Model Approach

- Mental Model approach (Johnson-Laird, 1999): A specific situation represented in a person's mind that can be used to help determine the validity of syllogisms in deductive reasoning
  - Create a model of a situation
  - Generate tentative conclusions about model
  - Look for exceptions to falsify model
  - Determine validity of syllogism
  - ➔ A conclusions is valid only if it cannot be refuted by any model of the premises

Deductive Reasoning

- Conditional syllogisms: have two premises and a conclusion
  - The first premise if of the form:"If p, then q"
  - There four major types of conditional syllogism

SYLLOGISM	SECOND PREMISE	CONCLUSION	IS IT VALID?	JUDGED CORRECTLY?
Syllogism 1: Modus ponens	*p*	Therefore, *q*	Yes	97%
Syllogism 2: Modus tollens	Not *q*	Therefore, not *p*	Yes	60%
Syllogism 3	*q*	Therefore, *p*	No	40%
Syllogism 4	Not *p*	Therefore, not *q*	No	40%

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Deductive Reasoning

- Conditional syllogisms: have two premises and a conclusion
  - The first premise if of the form:"If p, then q"
  - There four major types of conditional syllogism
  - Conditional Syllogism 1:
    - *If I study, I'll get a good grade.*
    - *I studied.*
    - *Therefore, I'll get a good grade.*



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Deductive Reasoning

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  - There four major types of conditional syllogism
  - Conditional Syllogism 2:
    - *If I study, I'll get a good grade.*
    - *I didn't get a good grade.*
    - *Therefore, I didn't study.*



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    - *If I study, I'll get a good grade.*
    - *I got a good grade.*
    - *Therefore, I studied.*



If I live in Tucson, then I live in Arizona.

I live in Arizona.

Therefore, I live in Tucson.

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    - *I didn't study.*
    - *Therefore, I didn't get a good grade.*

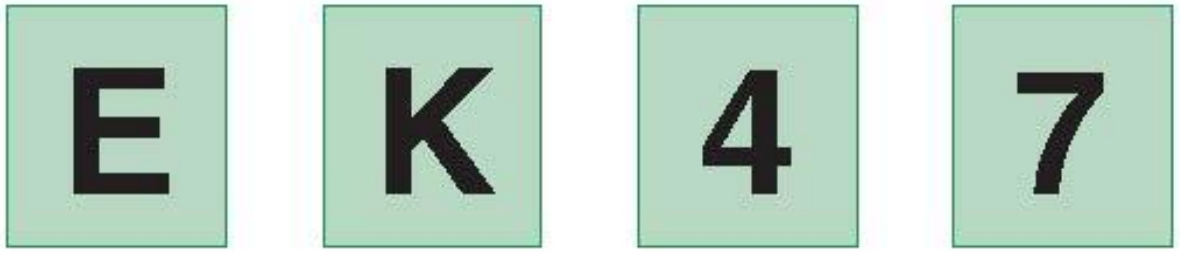


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The Wason Four-Card Problem

- Effect of using real-world items in a conditional-reasoning problem
  - Determine minimum number of cards to turn over to test: *if there is a vowel on one side, then there is an even number on the other side*



If vowel, then even number.

E needs to be turned over

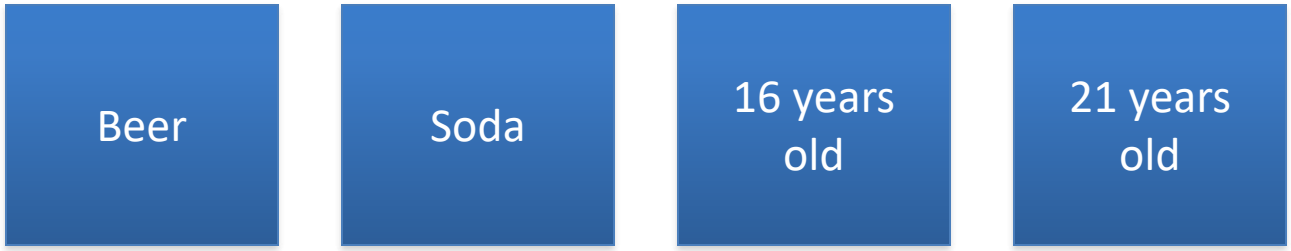
46% indicated that 4 would need to be turned over

Only 7% indicated that 7 needs to turned over

The Wason Four-Card Problem

- Falsification principle: to test a rule, you must look for situations that falsify the rule
  - Most participants fail to do this
  - When problem is stated in concrete everyday terms, correct responses greatly increase

If drinking beer, then 21 years old



73% of S got beer and 21 years old

The Wason Four-Card Problem

- Permission schema: if A is satisfied, B can be carried out
  - Used in the concrete versions
  - People are familiar with rules

- Context is important
- Familiarity is not always important